

● EnergyLens

Platform Showcase & Walkthrough

Microsoft Fabric-native energy intelligence for commercial buildings

Prepared for Dr. Kaddour Chelabi (BSBI) — mentor meeting

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Demo video: <https://youtu.be/OliqssHYtkY>

● WHY THIS MATTERS

Commercial buildings are the blind spot of the energy transition

40%

of EU energy is consumed by buildings

(residential + non-residential)

36%

of EU energy-related CO₂ emissions

the single largest sector

2030-2033

EU minimum performance deadlines

EPBD MEPS now binding

Yet most portfolios still run on monthly bills and spreadsheets

They are blind to anomalies, to regulatory risk (EPBD, EnEg, GEG, CSRD, the EU Battery Regulation), and to decarbonization opportunities. As EU rules tighten through 2033, owners face rising compliance exposure with no modern tooling — exactly the gap EnergyLens fills for the mid-market.

● THE PLATFORM IN ONE LINE

An energy intelligence & decision-support platform on Microsoft Fabric

It unifies multi-building energy data, scores it against EU regulation, and surfaces it through three layers:

● 1 Power BI report

9 analytical pages — consumption, anomalies, forecasting, sustainability (ESG/GHG), HVAC, real-time IoT, battery ROI.

● 2 Web application

Next.js + FastAPI + PostgreSQL. Navigation, access control, custom Fabric pages, PDF export, audit log.

● 3 AI Copilot

Answers questions over the Lakehouse using tool use — real queries, not text summaries — and can act.

On top: EU compliance scoring — CRREM decarbonization pathways, EnEfG, GEG, and CSRD / ESRS E1 reporting.

Medallion architecture on OneLake, three parallel paths into Fabric

SOURCES

Meters · bills · BMS sensors (BACnet/Modbus/MQTT) · weather (Open-Meteo) · prices (ENTSO-E) · grid CO₂

BRONZE

Raw, as ingested

SILVER

Cleaned, normalized, validated · single reference layer for factors & tariffs

GOLD

Business-ready KPIs, anomalies, simulations

Three validated paths to Fabric: Power BI DirectLake embed · custom React via SQL Analytics Endpoint · AI Copilot via LLM tool use

PART 1

The Power BI Report

Nine analytical pages reading the Gold layer through DirectLake mode

Page 1 · Portfolio Overview

How is my portfolio doing this month? (Energy Manager / C-suite)

[SCREEN RECORDING / SCREENSHOT]

Capture Page 1: the KPI cards (total kWh, cost €, CO₂), the per-building EUI heat map, and the worst-3 list.

WHAT IT SHOWS

Portfolio totals and a per-building energy-intensity comparison against type benchmarks, with the worst performers surfaced.

HOW IT'S CALCULATED

EUI = annual kWh ÷ conditioned m². Climate-adjusted by heating + cooling degree-days (EN ISO 15927-6) for fair comparison.

KEY TERM EUI — kWh/m²/year

Page 2 · Building Deep-Dive

What is this building doing right now? (Facility Manager)

[SCREEN RECORDING / SCREENSHOT]

Capture Page 2: the 24-hour hourly demand profile, today's peak vs baseline, and cost-to-date.

WHAT IT SHOWS

A single building's hourly demand curve, peak demand vs baseline, month-to-date cost and climate-adjusted EUI.

HOW IT'S CALCULATED

Hourly kW from gold_kpi_hourly. Peak = MAX(demand); base load = MIN at 02:00–04:00 (a high base load = equipment never sleeps).

KEY TERM Peak demand & base load

Page 3 · Anomaly Detection

Where is energy being wasted or a fault developing?

[SCREEN RECORDING / SCREENSHOT]

Capture Page 3: anomaly counts by severity, the type breakdown, and the table with estimated € loss.

WHAT IT SHOWS

Detected anomalies by severity and type, each with a probable cause, recommended action and an estimated euro loss.

HOW IT'S CALCULATED

One authoritative engine, 7 rule families (consumption spike, base load, COP drop, solar, battery, holiday, insulation), trilingual.

KEY TERM AFDD — fault detection

Page 4 · Forecasting

What will next month or quarter cost?

[SCREEN RECORDING / SCREENSHOT]

Capture Page 4: the 7/30/90-day forecast line with its confidence band, plus forecast cost and CO₂.

WHAT IT SHOWS

A per-building consumption forecast with a confidence band, translated into forecast cost and CO₂.

HOW IT'S CALCULATED

Meta's Prophet model, per building, with degree-days and occupancy as regressors; building-type-aware thresholds.

KEY TERM Prophet time-series forecast

Page 5 · Decision Support

What should I do, in what order, and what does it save?

[SCREEN RECORDING / SCREENSHOT]

Capture Page 5: ranked recommendations with €/kWh/CO₂ savings, effort, payback and matched incentives.

WHAT IT SHOWS

Actions ranked by a priority score, each with annual savings, effort, payback years and a matching grant (KfW/BAFA/YEKA/EEG).

HOW IT'S CALCULATED

Multi-country regulatory engine (DE/AT/NL/TR/EU) + incentive matching. $\text{priority} = (\text{saving} \times \text{urgency}) \div \text{effort}$.

KEY TERM Incentive matching

Page 6 · Sustainability (ESG / GHG)

What is my carbon footprint, am I report-ready? (the flagship reporting page)

[SCREEN RECORDING / SCREENSHOT]

Capture Page 6: the Scope 1/2/3 donut, the year-over-year bar, the area-weighted EPC map and the ESG score.

WHAT IT SHOWS

GHG emissions split into Scope 1/2/3, an area-weighted EPC compliance map, and a 0–100 ESG score. Maps to ESRS E1-5 / E1-6.

HOW IT'S CALCULATED

Scope 2 = grid kWh × sourced, year-indexed factor. EPC is area-weighted: $\Sigma(\text{area} \times \text{score}) \div \Sigma(\text{area})$ — not a plain average.

KEY TERM Scope 1/2/3 · ESRS E1

Page 7 · HVAC & Hourly Profile

Is my heating / cooling system efficient?

[SCREEN RECORDING / SCREENSHOT]

Capture Page 7: heat-pump COP (actual vs rated), the supply/return temperature delta, runtime and HVAC share.

WHAT IT SHOWS

Heat-pump performance vs its rating, supply/return delta-T as an efficiency proxy, and HVAC's share of consumption.

HOW IT'S CALCULATED

$\text{COP} = \text{heat produced} \div \text{electricity consumed}$, compared within equal outdoor-temperature bands so a cold-weather dip isn't a 'fault'.

KEY TERM COP / SCOP

Page 8 · Real-Time IoT Monitoring

What's happening now, which fault is burning money? (FLAGSHIP)

[SCREEN RECORDING / SCREENSHOT]

Capture Page 8: the 4 KPI cards, the 24h power trend (building vs HVAC), the sensor matrix and the FDD diagnostics.

WHAT IT SHOWS

Live power vs baseline, zone-comfort %, CO₂ level and automated fault diagnostics with an estimated daily euro cost.

HOW IT'S CALCULATED

Event Hub → EventStream → KQL Eventhouse + Lakehouse. ASHRAE Guideline 36 AFDD: 12 rules, frozen fault codes, ~31 sensor types.

KEY TERM ASHRAE Guideline 36

Page 9 · Battery Dispatch & ROI

Should I invest in a battery — which, and when does it pay back? (CFO / investor)

[SCREEN RECORDING / SCREENSHOT]

Capture Page 9: the chemistry × country matrix, the 7 strategies compared, and payback / NPV / IRR.

WHAT IT SHOWS

A financial comparison of battery chemistries and dispatch strategies, with payback, NPV, IRR and an EU-compliance flag.

HOW IT'S CALCULATED

NPV (5% discount, 10-yr, inflation + salvage), Newton-Raphson IRR, payback. EU Battery Regulation 2023/1542 compliance check.

KEY TERM NPV · IRR · payback

PART 2

The Web Application

Navigation, access control, custom Fabric pages, PDF export and audit logging

/portfolio

The home dashboard

[SCREEN RECORDING / SCREENSHOT]

Capture /portfolio: the portfolio + on-site-solar KPI rows, the buildings table and the Export-PDF button.

WHAT IT SHOWS

Portfolio-wide KPIs and a per-building table, with glossary tooltips, a 'data-not-connected' banner and PDF export.

HOW IT'S CALCULATED

Custom React reading the Fabric Lakehouse directly via the SQL Analytics Endpoint (ODBC) — one of three data paths.

KEY TERM SQL Analytics Endpoint

/buildings

List • detail • compare

[SCREEN RECORDING / SCREENSHOT]

Capture a building detail page: the embedded Power BI report, plus the list and compare views.

WHAT IT SHOWS

Building list (persistent selection, sticky compare bar) → detail with the embedded report → side-by-side compare.

HOW IT'S CALCULATED

Embedded via service principal (app-owns-data) so end users need no Power BI licence; module-locked pages show a 'locked' preview.

KEY TERM App-owns-data embed

/solar

Dedicated solar view

[SCREEN RECORDING / SCREENSHOT]

Capture /solar: the solar KPI cards and the generation / self-consumption visuals.

WHAT IT SHOWS

Self-consumption rate, self-sufficiency rate, yield and performance ratio for buildings with PV.

HOW IT'S CALCULATED

Self-consumption = $\text{used} \div \text{generated}$; self-sufficiency = $\text{solar} \div \text{total need}$; PR = $\text{actual} \div \text{theoretical yield}$.

KEY TERM Self-consumption vs sufficiency

/compliance

The EU-agenda hub

[SCREEN RECORDING / SCREENSHOT]

Capture /compliance: scroll the five panels and the Compliance-PDF export.

WHAT IT SHOWS

Five EU-regulation features in one hub — MEPS Radar, CRREM Stranding, Flexibility Readiness, EU Taxonomy, ESRS-E1 — each with an anti-greenwashing caveat.

HOW IT'S CALCULATED

Indicative banding (EPC + EUI) against EPBD 2030/2033, 1.5°C CRREM pathways, and ESRS E1-6 Scope 1/2/3 export.

KEY TERM Positions us vs Schneider / Siemens

/actions & /alerts

Recommendation workflow + anomaly triage

[SCREEN RECORDING / SCREENSHOT]

Capture /actions (status workflow) and /alerts (severity triage queue).

WHAT IT SHOWS

Recommendations and anomalies read from Fabric, with a Postgres status/acknowledge overlay and a nav badge.

HOW IT'S CALCULATED

'Unhandled' = unresolved (Fabric) AND not-acknowledged (Postgres); every change is written to the audit log.

KEY TERM Hybrid Fabric + Postgres

More modules

Glossary · Settings · Admin · Billing · Demo · PDF

[SCREEN RECORDING / SCREENSHOT]

Optional: quick screenshots of /glossary, /settings, /admin and the public /demo.

WHAT IT SHOWS

A glossary (tooltips app-wide), org & member settings, a founder-only admin console, Stripe self-serve billing, and a public demo URL.

HOW IT'S CALCULATED

Stripe billing · audit log · 4 PDF report routes · graceful degradation · mobile + accessibility.

KEY TERM Production-grade polish

PART 3

The AI Copilot

Conversational analytics over the Lakehouse — using tool use, not text summaries

It runs real queries — and it can act, not just describe

Ask in plain language; the model picks a tool, the dispatcher queries Fabric/Postgres, the answer streams back.

[SCREEN RECORDING / SCREENSHOT]

Capture /copilot: type a question, show the tool call happening, and the streamed answer.

SIX PRODUCTION TOOLS

query_kpi	KPIs for a building
compare_buildings	rank / compare buildings
list_recommendations	recommendations by priority
get_anomalies	anomalies by severity/type
simulate_battery_scenario	payback / NPV / IRR
update_action_status	change status (a write)

Provider abstraction: Anthropic / Azure OpenAI / Mock — production is one environment variable. RLS enforced on every tool call.

PART 4

Market & Strategy

Why it matters, how big it gets, and how we win

● MARKET SIZE

A large, growing market — not a niche in decline

€18B

TAM — EU commercial buildings

~5M buildings × €100–700/mo

€4.3B

SAM — DACH focus

highest-value compliance market

**\$8.4→2
4B**

BEMS software market

2026 → 2035

8.7%

BEMS market CAGR

to 2035

Five tailwinds, all at once

Binding EU climate regulation (2024–2033) · the post-2022 energy-price shock made energy a board topic · Microsoft Fabric matured (GA 2024) · IoT sensor costs collapsed · AI made pattern recognition affordable. The reasons to buy are structural and accelerating.

Microsoft-native + EU-regulation niche + mid-market — a near-empty intersection

PLAYER	WHERE THEY ARE	GAP ENERGYLENS FILLS
Measurabl / Aquicore	US enterprise, well-funded	Mid-market EU coverage
Cortexa	EU enterprise	Mid-market accessibility & price
Schneider EcoStruxure	Peer: auditable reporting + AI	We add multi-protocol IoT + Fabric-native
Siemens Desigo	BMS / building automation	We are the analytics layer, not a BMS

Our angle: an AI Copilot that reasons over the Lakehouse via tool use — under €299/building/month.

Mid-market SaaS economics that work for a lean team

Insight

€99

/building/month

SME offices

Monitor

€299

/building/month

mid-market property mgmt

Copilot

€699+

/building/month

enterprise / IoT-equipped

Custom

€5-50k

/building/month

10+ building portfolios

36-108×

LTV / CAC

2-4 mo

payback period

~85%

gross margin

First three months free (pilot) to win reference cases • break-even at customer #2.

Bottom-up, validation-first — then expand

Now → 6 mo

Free university & DACH mid-market pilots (close in 2–8 weeks). BSBI campus pilot first.

6–18 mo

Convert pilots to paying customers; case studies open hotels, healthcare, REITs.

18 mo+

Microsoft Partner Network co-selling; expand DACH → BeNeLux → Southern Europe.

The 3-year vision: the default Microsoft Fabric-native energy-intelligence platform for European commercial real estate — 100+ buildings, €1–2M ARR, a Microsoft partner in climate-tech.

● WHY THIS FOUNDER

Energy domain depth × modern data engineering — a rare combination

5 weeks

to build the whole stack, solo

pipeline → app → AI Copilot

DP-600

Microsoft Fabric certified

few solo founders hold it

B.Sc.

+M.Sc.
Energy Engineering + Management

Yaşar + BSBI (June 2026)

Most BI developers don't know kWh/m² or Scope 2; most energy engineers don't know medallion or tool use. EnergyLens sits at the intersection — and was executed end-to-end, solo, accelerated by AI-assisted development. An 18-month post-graduation visa gives the runway to incorporate in Germany.

THE ASK

How you can help

1 · EXIST academic mentor

EXIST is the German federal founder grant (~€65k / 12 months). It requires an academic mentor letter. Since EnergyLens grew from my M.Sc. work, I'd be honored if you'd serve as mentor — one short letter (I'll draft it) and a couple of alignment chats.

2 · BSBI campus pilot

A free, six-month, RLS-secured pilot I run end-to-end — real-time visibility for the facilities team, a named case study for BSBI, and validation for my M.Sc. capstone. No cost, no commitment.

3 · Ongoing advisory (optional)

Occasional monthly / quarterly check-ins on the energy-management and academic angles. No formal commitment — just openness to input.

● Thank you.

I'd welcome your guidance — and your support on the EXIST application.

Ali Mert Özdemir · energylens.eu · Microsoft DP-600

Demo: <https://youtu.be/0liqssHYtkY>